

Modelling and analysis of panel count data with multiple modes of recurrence

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Abstract

Survival studies commonly address two distinct types of data: recurrent event data and panel count data. Specifically, some of the study individuals may be continuously monitored throughout the study period, while others may only be evaluated at discrete time intervals. Panel count data refers to the information collected in studies focusing on recurrent events, where subjects are observed only at discrete time points. If these study subjects are exposed to recurrent events of several types, we obtain panel count data with multiple modes of recurrence. We introduce cause specific mean function and study its properties. Non-parametric estimation of the cause specific mean functions is presented using isotonic regression method. We introduce proportional cause specific mean model to study the effect of covariates on recurrence time. The estimation of regression parameter and baseline cause specific mean functions are discussed. Simulation studies are carried out to evaluate the finite sample performance of the proposed estimators. The procedures are applied to two real data sets, to demonstrate the practical utility..